Relativistic Quantum Field Theory I(PH.60053)
Exercise Set 1

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Note: I generally disdain the use of natural units in undergraduate studies. For students having gone through a symbolic mess of a calculation, something as simple as dimensional analysis gives a good sanity check. And while this course is explicitly designed for graduate students, I would like to stand by this conviction and perform all calculations with explicit engineering dimensions and constants. I believe it is reasonably simple for the grader to convert it into natural units. I appreciate your understanding.

Problem 1

We use the following definitions:

$$A^\mu = (\frac{\phi}{c}, \vec{A}), \quad F^{\mu\nu} = \partial^{[\mu}A^{\nu]}, \quad J^\mu = (-c\rho, \vec{J}), \quad \partial_\mu = (\frac{1}{c}\partial_t, \vec{\nabla}), \quad \partial^\mu = (-\frac{1}{c}\partial - t, \vec{\nabla}),$$

$$E^i = -\partial_i\phi - \partial_t A^i = c\left(\left(-\frac{1}{c}\partial_t\right)A^i - \partial_i\left(\frac{\phi}{c}\right)\right) = cF^{0i},$$

$$B^i = \partial_j A^k - \partial_k A^j = F^{jk} \quad (\epsilon_{ijk} = +1).$$

We start with the Lagrangian density:

$$\mathcal{L} = -\frac{1}{4\mu_0}F_{\alpha\beta}F^{\alpha\beta} + J_\alpha A^\alpha.$$

We find the following partial derivative relations:

$$\frac{\partial(\partial_\alpha A_\beta)}{\partial(\partial_\mu A_\nu)} = \delta^\mu_\alpha\delta^\nu_\beta \Rightarrow \frac{\partial(\partial^\alpha A^\beta)}{\partial(\partial_\mu A_\nu)} = \eta^{\mu\alpha}\eta^{\nu\beta}$$

$$\Rightarrow \frac{\partial}{\partial(\partial_\mu A_\nu)}(F_{\alpha\beta}F^{\alpha\beta}) = (\delta^\mu_\alpha\delta^\nu_\beta - \delta^\mu_\beta\delta^\nu_\alpha)F^{\alpha\beta} + F_{\alpha\beta}(\eta^{\mu\alpha}\eta^{\nu\beta} - \eta^{\mu\beta}\eta^{\nu\alpha}) = 4F^{\mu\nu}$$

Thus, the Euler-Lagrange equations are given by

$$0 = \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu A_\nu)}\right) - \frac{\partial\mathcal{L}}{\partial A_\nu} = \partial_\mu\left(-\frac{1}{\mu_0}F^{\mu\nu}\right) - J^\nu$$

$$\therefore \partial_\mu F^{\mu\nu} = \mu_0 J^\nu$$

1. $\nu = 0$: $\partial_\mu F^{\mu 0} = -\frac{1}{c}\vec{\nabla}\cdot\vec{E} \Rightarrow \vec{\nabla}\cdot\vec{E} = \frac{\rho}{\epsilon_0}$

2. $\nu = 1$: $\partial_\mu F^{\mu 1} = \left(\frac{1}{c}\partial_t\right)\left(\frac{E_x}{c}\right) + \partial_y(-B^z) + \partial_z B^y = \left(\frac{1}{c^2}\partial_t \vec{E} - \vec{\nabla} \times \vec{B}\right)^x$.
   Thus, generally, $\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \frac{1}{c^2}\partial_t \vec{E}$.

Furthermore, the Bianchi identity can be expressed as follows:

1. $\partial_{[0}F_{12]} = \frac{1}{c}\partial_t B^z + \partial_x\left(+\frac{E^y}{c}\right) + \partial_y\left(-\frac{E^x}{c}\right) = \frac{1}{c}\left(\partial_t \vec{B} + \vec{\nabla} \times \vec{E}\right)^z$.
   Thus, generally, $\vec{\nabla} \times \vec{E} = -\partial_t \vec{B}$.
2. $\partial_{[1}F_{23]} = \partial_x B^x + \partial_y B^y + \partial_z B^z = \vec{\nabla} \cdot \vec{B} = 0$.

Thus, we see that the Euler-Lagrange equations as well as the Bianchi identity successfully reproduce the conventional Maxwell equations. $\square$

Problem 2

From the special-relativistic identity $E^2 = \sqrt{\vec{p}^2 c^2 + m^2 c^4}$, we attempt to define the Hamiltonian operator as:

$$H \stackrel{?}{=} \sqrt{-\hbar^2 c^2 \vec{\nabla}^2 + m^2 c^4} = mc^2 \sqrt{1 - \left(\frac{\hbar}{mc}\vec{\nabla}\right)^2}.$$

To transform this into a kernel function, i.e., find the function $K(\vec{x}, \vec{y})$ such that

$$(H\psi)(\vec{x}) = \int d^3\vec{y} K(\vec{x}, \vec{y})\psi(\vec{y}),$$

we utilize the Fourier transform:

$$\psi(\vec{x}) = \int d^3\vec{k}\psi_{\vec{k}} e^{i\vec{k}\cdot\vec{x}}, \quad \psi_{\vec{k}} = \int \frac{d^3\vec{x}}{(2\pi)^3}\psi(\vec{x})e^{-i\vec{k}\cdot\vec{x}}.$$

Substitution then yields:

$$\begin{aligned}
(H\psi)(\vec{x}) &= mc^2 \int d^3\vec{k}\psi_{\vec{k}} \sqrt{1 - \left(\frac{\hbar}{mc}\vec{\nabla}\right)^2} e^{i\vec{k}\cdot\vec{x}} \\
&= mc^2 \int d^3\vec{k}\psi_{\vec{k}} \sqrt{1 - \left(\frac{\hbar\vec{k}}{mc}\right)^2} e^{i\vec{k}\cdot\vec{x}} \\
&= mc^2 \int d^3\vec{k} \left(\int \frac{d^3\vec{y}}{(2\pi)^3}\psi(\vec{y})e^{-i\vec{k}\cdot\vec{y}}\right) \sqrt{1 - \left(\frac{\hbar\vec{k}}{mc}\right)^2} e^{i\vec{k}\cdot\vec{x}} \\
&= \int d^3\vec{y} \left(mc^2 \int \frac{d^3\vec{k}}{(2\pi)^3} \sqrt{1 - \left(\frac{\hbar\vec{k}}{mc}\right)^2} e^{i\vec{k}\cdot(\vec{x}-\vec{y})} \right) \psi(\vec{y})
\end{aligned}$$

Thus, defining $\vec{r} := \vec{x} - \vec{y}$, we find that

$$\begin{aligned}
K(\vec{x},\vec{y}) &= mc^2 \int \frac{d^3\vec{k}}{(2\pi)^3} \sqrt{1 - \left(\frac{\hbar\vec{k}}{mc}\right)^2} e^{i\vec{k}\cdot\vec{r}} \\
&= \frac{m^4c^5}{8\pi^3\hbar^3} \int d^3\vec{\kappa} \sqrt{1+\vec{\kappa}^2} e^{i\vec{\kappa}\cdot\vec{\rho}} \qquad \left(\vec{\kappa} := \frac{\hbar\vec{k}}{mc}, \quad \vec{\rho} := \frac{mc\vec{r}}{\hbar}\right) \\
&= \frac{m^4c^5}{8\pi^3\hbar^3} \int_0^\infty d\kappa\kappa^2 \sqrt{1+\kappa^2} \int_{-1}^1 d(\cos\theta) e^{i\kappa\rho\cos\theta} \int_0^{2\pi} d\phi \\
&= \frac{m^4c^5}{2\pi^2\hbar^3\rho} \int_0^\infty d\kappa\kappa\sqrt{1+\kappa^2}\sin\kappa\rho.
\end{aligned}$$

Despite the obvious problem of convergence, we nevertheless attempt to sidestep the issue using appropriate Feynmann tricks.

$$\begin{aligned}
K(\vec{x},\vec{y}) &= \frac{m^4c^5}{2\pi^2\hbar^3\rho} \int_0^\infty d\kappa \left(\frac{\kappa^3}{\sqrt{1+\kappa^2}} + \frac{\kappa}{\sqrt{1+\kappa^2}}\right) \sin\kappa\rho \\
&= \frac{m^4c^5}{2\pi^2\hbar^3\rho} \left(\frac{d^3}{d\rho^3} - \frac{d}{d\rho}\right) \int_0^\infty d\kappa \frac{\sin\kappa\rho}{\sqrt{1+\kappa^2}} \\
&= \frac{m^4c^5}{2\pi^2\hbar^3\rho} \left(K_0'''(\rho) - K_0'(\rho)\right),
\end{aligned}$$

where $K_0(x)$ is the modified Bessel function of the second kind of order 0. The last expression can be simplified using well-known recurrence relations:

$$K_0'(x) = -K_1(x), \quad K_1'(x) = -K_0(x) - \frac{1}{x}K_1(x), \quad K_2(x) = K_0(x) + \frac{2}{x}K_1(x).$$

$$\Rightarrow K_0'''(x) - K_0'(x) = -\frac{1}{x}K_2(x)$$

$$\therefore K(\vec{x},\vec{y}) = -\frac{m^4c^5}{2\pi^2\hbar^3}\frac{K_2(\rho)}{\rho^2} = -\frac{m^2c^3}{2\pi^2\hbar}\frac{K_2\left(\frac{mc}{\hbar}r\right)}{r^2}$$

Notice the blatant nonlocality: $K(\vec{x},\vec{y})$ has support arbitrarily far away from $\vec{x}=\vec{y}$! In fact, the asymptotic behavior $K_\alpha(x) \overset{x\to\infty}{\sim} \sqrt{\frac{\pi}{2x}}e^{-x}$, we find that

$$K(\vec{x},\vec{y}) \overset{r\to\infty}{\sim} -\sqrt{\frac{m^3c^5}{8\pi^3\hbar}} r^{-2} e^{-\frac{mc}{\hbar}r}.$$

Problem 3

We first need to reconstruct the omitted natural constants in the formulae. With the ground truths

$$[\mathcal{L}] = [EL^{-3}] \quad \text{and} \quad [\partial] = [L^{-1}],$$

we insist that

$$[\mathcal{L}] = [\partial^\mu\phi\partial_\mu\phi] = [\phi^2 \cdot L^{-2}] \Rightarrow [\phi] = [E^{1/2}L^{-1/2}].$$

(Note: I could not decide whether this is the most useful convention. I also think it would be reasonable to insist that ϕ is dimensionless and multiply by appropriate natural constants to match the dimensions. This is just the convention that Gemini suggested, so it's its fault.) Moreover, we insist that $[m] = [M] = [EL^{-2}T^2]$, which implies

$$[m^2\phi^2] = [(EL^{-2}T^2)^2 \cdot EL^{-1}] = [E^3L^{-5}T^4] = [EL^{-3} \cdot (ET)^2 \cdot (LT^{-1})^{-2}] = [\mathcal{L} \cdot \hbar^2 c^{-2}].$$

Thus, the reconstructed Lagrangian density is given by

$$\mathcal{L} = \frac{1}{2}\left(-\partial_\mu\phi\partial^\mu\phi - \left(\frac{mc}{\hbar}\right)^2\phi^2\right).$$

Now, we can actually start solving the problem. The kinetic term can be expanded as:

$$\mathcal{L} = \frac{1}{2}\left(\left(\frac{\dot{\phi}}{c}\right)^2 - \left(\vec{\nabla}\phi\right)^2 - \left(\frac{mc}{\hbar}\right)^2\phi^2\right),$$

which lets us find the conjugate momentum field:

$$\pi(t,\vec{x}) := \frac{\partial\mathcal{L}}{\partial\dot{\phi}(t,\vec{x})} = \frac{\dot{\phi}}{c^2}.$$

Thus, the Hamiltonian functional is given by:

$$\begin{aligned} H[\phi,\pi;t] &= \int d^3\vec{x}\left(\pi(t,\vec{x})\dot{\phi}(t,\vec{x}) - \mathcal{L}[\phi,\dot{\phi}]\right) \\ &= \int d^3\vec{x}\frac{1}{2}\left((c\pi(t,\vec{x}))^2 + \left(\vec{\nabla}\phi(t,\vec{x})\right)^2 + \left(\frac{mc}{\hbar}\phi(t,\vec{x})\right)^2\right). \end{aligned}$$

We now write down the Heisenber equations of motion:

$$\frac{dA}{dt} = -\frac{1}{i\hbar}\left[H[\phi,\pi;t],A\right].$$

The equation for $\phi(t,\vec{x})$ is relatively straightforward:

- $\left[\pi(t,\vec{y})^2,\phi(t,\vec{x})\right] = \{\pi(t,\vec{y}),[\pi(t,\vec{y}),\phi(t,\vec{x})]\} = -2i\hbar\delta^3(\vec{y}-\vec{x})\pi(t,\vec{y})$
- $\left[\vec{\nabla}_y^2\phi(t,\vec{y}),\phi(t,\vec{x})\right] = \vec{\nabla}_y^2\left[\phi(t,\vec{y}),\phi(t,\vec{x})\right] = 0$
- $\left[\phi(t,\vec{y})^2,\phi(t,\vec{x})\right] = \{\phi(t,\vec{y}),[\phi(t,\vec{y}),\phi(t,\vec{x})]\} = 0$

$$\therefore \dot{\phi}(t,\vec{x}) = -\frac{1}{i\hbar}\left[H[\phi,\pi;t],\phi(t,\vec{x})\right] = -\frac{1}{i\hbar}\int d^3\vec{y}\left(-i\hbar c^2\delta^3(\vec{y}-\vec{x})\pi(t,\vec{y})\right) = c^2\pi(t,\vec{x})$$

The commutator relation with $\pi(t,\vec{x})$ is a bit more tedious:

- $\left[\pi(t,\vec{y})^2,\pi(t,\vec{x})\right] = \{\pi(t,\vec{y}),[\pi(t,\vec{y}),\pi(t,\vec{x})]\} = 0$
- $\left[\phi(t,\vec{y})^2,\pi(t,\vec{x})\right] = \{\phi(t,\vec{y}),[\phi(t,\vec{y}),\pi(t,\vec{x})]\} = 2i\hbar\delta^3(\vec{y}-\vec{x})\phi(t,\vec{y})$

- $\left[\int d^3\vec{y}\left(\vec{\nabla}_y\phi(t,\vec{y})\right)^2, \pi(t,\vec{x})\right]$

$$
\begin{aligned}
&= \left[\oint_{bdry} d^2\vec{S}_y\cdot\left(\phi(t,\vec{y})\vec{\nabla}_y\phi(t,\vec{y})\right) - \int d^3\vec{y}\phi(t,\vec{y})\vec{\nabla}_y^2\phi(t,\vec{y}), \pi(t,\vec{x})\right] \\
&= -\int d^3\vec{y}\left(\phi(t,\vec{y})\vec{\nabla}_y^2\left[\phi(t,\vec{y}),\pi(t,\vec{x})\right] + \left[\phi(t,\vec{y}),\pi(t,\vec{x})\right]\cdot\vec{\nabla}_y^2\phi(t,\vec{y})\right) \\
&= -i\hbar\int d^3\vec{y}\left(\phi(t,\vec{y})\vec{\nabla}_y^2\delta^3(\vec{y}-\vec{x}) + \delta^3(\vec{y}-\vec{x})\vec{\nabla}_y^2\phi(t,\vec{y})\right) \\
&= -i\hbar\left(\oint_{bdry} d^2\vec{S}_y\cdot\left(\phi(t,\vec{y})\vec{\nabla}_y\delta^3(\vec{y}-\vec{x})\right) - \int d^3\vec{y}\vec{\nabla}_y\phi(t,\vec{y})\cdot\vec{\nabla}_y\delta^3(\vec{y}-\vec{x}) + \vec{\nabla}^2\phi(t,\vec{x})\right) \\
&= i\hbar\left(\int d^3\vec{y}\vec{\nabla}_y\phi(t,\vec{y})\cdot\vec{\nabla}_y\delta^3(\vec{y}-\vec{x}) - \vec{\nabla}^2\phi(t,\vec{x})\right) \\
&= i\hbar\left(\oint_{bdry} d^2\vec{S}_y\cdot\left(\delta^3(\vec{y}-\vec{x})\vec{\nabla}_y\pi(t,\vec{y})\right) - \int d^3\vec{y}\delta^3(t,\vec{y})\cdot\vec{\nabla}_y^2\phi(t,\vec{y}) - \vec{\nabla}^2\phi(t,\vec{x})\right) \\
&= -2i\hbar\vec{\nabla}^2\phi(t,\vec{x})
\end{aligned}
$$

$$
\begin{aligned}
\therefore \dot{\phi}(t,\vec{x}) &= -\frac{1}{i\hbar}\left[H[\phi,\pi;t],\pi(t,\vec{x})\right] \\
&= -\frac{1}{i\hbar}\left(\int d^3\vec{y}\frac{1}{2}\left(\frac{mc}{\hbar}\right)^2\cdot 2i\hbar\delta^3(\vec{y}-\vec{x})\phi(t,\vec{y}) + \frac{1}{2}\left(-2i\hbar\vec{\nabla}^2\phi(t,\vec{x})\right)\right) \\
&= \left(\vec{\nabla}^2 - \left(\frac{mc}{\hbar}\right)^2\right)\phi(t,\vec{x})
\end{aligned}
$$

Combining these two, we recover the Klein-Gordon equation:

$$\left(\partial_\mu\partial^\mu - \left(\frac{mc}{\hbar}\right)^2\right)\phi(t,\vec{x}) = 0.$$